

Bhaskar Ruthvik Bikkina

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EDUCATION

Texas A&M University

Master of Science in Computer Science - GPA: 4.0/4.0

August 2025 - May 2027

College Station, TX

Coursework: Operating Systems, Supercomputing, Algorithms, Large-scale ML Optimization, Information Retrieval

Birla Institute of Technology and Science Pilani

Bachelor of Engineering in Computer Science - GPA: 9.09/10

October 2021 - July 2025

Telangana, India

Coursework: Data Structures, Object Oriented Programming, Databases, Networks, Computer Architecture

TECHNICAL SKILLS AND CO-CURRICULARS

Languages: Python, C/C++, Java, Go, SQL, TypeScript, JavaScript, HTML/CSS, Bash

Frameworks and Tools: PostgreSQL, Kubernetes, Docker, AWS, GCP, FastAPI, Linux, OpenMP, CUDA, SLURM, Temporal, protobufs, PyTorch, vLLM, Grafana, CI/CD, Git, Unit Testing

Co-Curriculars: *Lead:* ACM BITS App Development, *Founder:* Open-Source Club, *Teaching Assistant:*

Data Analytics Platforms (TAMU), Data Mining (BITS), *Winner:* FD Innovation Day - Best Poster, BITS Hyderabad

EXPERIENCE

MTS Intern

zaimler, Inc.

May 2026 - August 2026

Full-time, San Mateo, CA

- Designed and built an API over long-running jobs on FastAPI and Temporal, polling progress to the client.
- Benchmarked candidate models on **Baseten** across quantization and speculative decoding, cutting latency **3-4x**.
- Upgraded a production service across **Docker** and **Kubernetes**, patching **Trivy**-flagged CVEs in the image.
- Benchmarked each patched image against the incumbent deployment, confirming no inference performance drop.
- Prototyped compiling an IR to **Cypher** in an agentic workflow, constraining generation over free-form drafting.
- Used **Claude Code** for incremental development, reviewing all generated code before merge to production.

Data Science Intern

American Express

July 2024 - December 2024

Full-time, Gurugram, India

- Pioneered a novel approach to unify **XGBoost** regression trees in **Python** with a performance drop less than 1%.
- Performed extensive case studies using **SQL**, **Spark** to gain insights, engineer features, alter boosting process.
- Utilized Bayesian Optimization and improved gini evaluation metric by **33%**, beating standalone models by **14%**.

Undergraduate Research Assistant

Birla Institute of Technology and Science Pilani

March 2024 - June 2025

Part-time, Hyderabad, India

- Built a retrieval system over a **FAISS** vector index, serving semantic search results via a Flask backend.
- Improved retrieval quality by **53%** and reduced inference latency by **30%** using in-context learning.
- Research papers based on this project were accepted by **ECIR 2025** and HCI+NLP Workshop at EMNLP 2025.

PROJECTS

Barebones Operating System | C++, x86 Kernel, Assembly, QEMU, gdb

January 2026 - April 2026

- Implemented two-level hierarchical paging with a page fault handler, relocating page tables into virtual memory.
- Added preemptive round-robin scheduling on an end-of-quantum timer, masking interrupts in critical sections.
- Replaced busy-wait disk I/O with a driver that blocks the thread and yields the CPU until the disk is ready.
- Built a file system over the block device with an inode table, free-list bitmap and sequential file read/write.

Pascal Compiler Implementation | C, Yacc, Lex

March 2024 - May 2024

- Implemented a bare-bones Pascal language compiler with loops, data structures, variables and conditionals.
- Utilized standard compiler implementation process using 3AC codes to help process the provided input.

Graph Database - Distributed Systems Architecture | C, Posix API

October 2023 - December 2023

- Implemented read, write and search functions for a simple graph database with a load balancer for scalability.
- Optimized the search process by utilizing multi-threading to distribute the workload and speed up retrieval.
- Utilized Shared Memory as a common data store for all threads and added semaphores to prevent race conditions.